MIL-A-722A
27 October 1961
SUPERSEDING
JAN-M-722
27 January 1949

MILITARY SPECIFICATION

ANALYZER, POWER CIRCUIT, AC, PORTABLE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. - This specification covers portable, alternating current power circuit analyzers capable of measuring the voltage, current, and power in single and polyphase circuits and power factor in balanced three phase, three wire alternating current circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION
Publication C39.1-1959 - American Standard for Electrical Indicating Instruments.

(Copies may be obtained from the American Institute of Electrical Engineers, 33 West 39th Street, New York 18, New York.)

3. REQUIREMENTS

3.1 Preproduction sample. - Preproduction samples of power circuit analyzing meters as specified (see 6.2) shall be furnished for preproduction inspection.

3.2 Definitions. - Definitions listed under American Standards Association Publication C39.1 - 1959 are applicable to the various technical terms which appear in this specification.

3.3 Basic requirements. - Instruments comprising the analyzers shall conform to A.S.A. publication C39.1 - 1959 except as specified herein.

3.4 Materials. - Materials which are capable of producing dangerous volumes of gases or other harmful toxic effects under operating conditions specified herein, including fire are prohibited for use in construction of the analyzer.

3.5 Design. -

3.5.1 Scale ranges. - Scale ranges for the instruments comprising the analyzer shall be as follows:

(a) Ammeter - 125/25/5 amperes
(b) Voltmeter - 600/300/150 volts
(c) Wattmeter - 0 - 100 kilowatts at 600 volts 125 amperes
(d) Power factor meter - 0.30 lag to 0.80 lead

3.5.2 Scale. - The instruments comprising the analyzer shall be 90 degree nominal scale, 4-1/2 inch rectangular alternating current electrical indicating instruments.

3.5.3 Accuracy rating. - Accuracy of the instruments comprising the analyzer shall be as follows:

(a) Ammeter - 1 percent of full scale value
(b) Voltmeter - 1 percent of full scale value
(c) Wattmeter - 1 percent of full scale value
(d) Power factor meter - 1 percent 0.01 power factor

3.5.4 Nominal frequency. - Instruments shall be designed for operation at 60 cycles per second.

FSC 6625

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3.5.5 Terminals. - Analyzer terminals shall be suitably insulated and located away from the operator.

3.5.6 Pointer. - Each instrument comprising the analyzer shall have pointers of the knife-edge type.

3.5.7 Pointer stop. - Ammeters shall be provided with an adjustable pointer stop to facilitate determination of the maximum value of motor starting currents.

3.5.8 Parallax. - Instruments shall either have a mirror on the dial near the scale and directly beneath the knife edge portion of the pointer for avoiding errors in reading due to parallax or the clearance between the pointer and the scale shall not exceed 0.10 inch nor be less than 0.05 inch.

3.5.9 Case. - The analyzer shall be enclosed in a box of seasoned hardwood with a hinged removable cover. The case shall be provided with durable rubber supporting feet and a suitable leather carrying handle. Corrosion resistant metal reinforcements shall be provided at each corner of the hardwood case.

3.5.10 Normal operating position. - Instruments shall be mounted on a horizontal panel within the analyzer enclosure.

3.5.11 Weight and dimensions. - The weight of the analyzer shall not exceed 37 pounds. The cubic content of the analyzer shall not exceed 2200 cubic inches.

3.5.12 Operating instructions. - Operating instructions shall be provided with each analyzer and shall be located on the inside surface of the removable cover of the case. Operating instructions shall be protected by a transparent cover of vinyl chloride plastic.

3.5.13 Nameplate. - The analyzer shall be provided with a permanently marked nameplate located on the instrument panel. The following minimum information shall be provided:

Standard nomenclature of item
Specification number
Name of manufacturer
Manufacturer's type and model number

3.6 Position influence. - When tested in accordance with 4.5.2 the pointer tip of the spring-controlled meters shall not deviate from the zero mark more than the amount specified (see 3.18).

3.7 Initial accuracy. - The initial or rated accuracy shall be within the limits specified (see 3.18) when tested in accordance with 4.5.3.

3.8 Damping factor. - When tested in accordance with 4.5.4 the damping factor shall be not less than the values specified (see 3.18).

3.9 Response time. - When tested in accordance with 4.5.5 the response time shall not exceed the values specified (see 3.18).

3.10 Power loss. - The power loss shall not exceed the values specified (see 3.18 and 4.5.6).

3.11 Temperature influence. - The temperature influence shall not exceed the values specified (see 3.18 and 4.5.7).

3.12 Extreme temperatures. - When tested in accordance with 4.5.9 the change in indication shall not exceed the values specified in 4.5.9.

3.13 Frequency influence. - When tested in accordance with 4.5.8 the change in indication shall not exceed the values specified (see 3.18).

3.14 Overload. -

3.14.1 Momentary. - The mechanism of instruments having current circuits shall be sufficiently rugged to withstand without mechanical or thermal injury the overload specified in 4.5.10.1. It shall not be necessary to make any repairs or adjustments other than resetting the pointer to zero by the zero adjuster. The permanent change in indication shall not exceed the rated accuracy of the instrument (see 3.18).

3.14.2 Sustained overload. - When tested in accordance with 4.5.10.2 the temporary and permanent zero shift and the permanent change in indication shall not exceed the values specified in 4.5.10.2.

3.15 Humidity resistance. - After the test specified in 4.5.11 instruments shall operate freely and the permanent change in indication shall not exceed the value specified in 4.5.11.

3.16 Dielectric stress. - The analyzer shall be capable of withstanding the test voltage specified in 4.5.12 without damage or flashover.

3.17 Insulation resistance. - When tested in accordance with 4.5.13 the insulation resistance shall be not less than 20 megohms.

3.18 Detailed requirements for individual meters. - All requirements specified herein are applicable to the instruments covered by Plates 16, 18, and 20 of A.S.A. publication C39.1 - 1959. Detailed requirements or exceptions applicable to the individual instruments are specified on plates 16, 18, and 20 which are listed in supplement to this

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specification. In the event of any conflict between the requirements of this specification and Plates 16, 18, and 20, the former shall govern.

3.19 Reference temperature. - The standard temperature of reference shall be 25°C ±5°C.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Preproduction samples shall be submitted to a laboratory acceptable to the Government. These samples shall be subjected to examination and tests specified in 4.5.1 to 4.5.13. Preproduction inspection may be waived as follows (see 6.2):

4.3 Sampling. -

4.3.1 Inspection lot. - All analyzers of the same type presented at one time shall be considered a lot for purposes of acceptance inspection.

4.3.2 Sampling for acceptance inspection. - A random sample of analyzers shall be selected in accordance with table I from each inspection lot offered for Government inspection in accordance with 4.4. Lot acceptance will be based on sampling inspection requirements in accordance with Standard MIL-STD-105.

Table I - Sampling for acceptance inspection
AQL (approx.) = 1.5 percent defective.

Number of analyzer in inspection lot	Number of analyzers in sample	Acceptance number (defective)	Rejection number (defective)
8 or under	2	0	1
9 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 and over	15	0	1

4.4 Testing for lot acceptance. - Each of the sample analyzers selected in accordance with table I shall be subjected to the examinations and tests specified in 4.5.1, 4.5.2, 4.5.3, 4.5.4, 4.5.5, 4.5.12, 4.5.13. Any analyzer in the sample which does not meet the requirements specified in 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.16, and 3.17 shall be rejected and the lot represented by the sample shall be rejected.

4.5 Examination and test procedures. -

4.5.1 Visual examination. - Analyzers shall be carefully examined to determine compliance with 3.3, 3.4, and 3.5, inclusive.

4.5.2 Position influence. - Pointers shall be set to the center of the zero scale division with the analyzer in the normal operating position. The analyzer shall then be tilted 60 degrees from this position in any direction, and the deviation of the pointers noted (see 3.6).

4.5.3 Initial accuracy. - The analyzer shall be checked for accuracy as specified in 4.5.3.1 or 4.5.3.2, as applicable.

4.5.3.1 Preproduction test. - After preliminary conditioning in the test circuit for 30 minutes at two-thirds scale deflection, readings shall be taken and compared with corresponding readings of standard instruments. The test shall be made with the analyzer at a temperature of 25°C. Readings shall be taken at six approximately equidistant points on the scale including the zero and end scale readings (see 3.7).

4.5.3.2 Inspection test. - Same as 4.5.3.1 except no preliminary conditioning required.

4.5.4 Damping factor. - Voltage or current sufficient to produce momentary full-scale deflection shall be applied to the analyzer. The maximum momentary deflection of each instrument shall be taken and recorded. After the pointer of each instrument has come to rest, the steady deflection shall be taken and recorded. (When making this test, the circuit resistance shall be sufficiently high so that no additional damping factors shall be taken as the ratio of the steady deflection in angular degrees to the difference between the maximum momentary deflection and the steady deflection in angular degrees.)

4.5.5 Response time. - Steady electric power sufficient to produce momentary end-scale deflection shall be applied to the analyzer. The length of time, in seconds, required for the pointer of each instrument to come to apparent rest shall be recorded. The pointer of each instrument shall be considered as having come to apparent rest when it has obtained the actual point of rest within one-half the initial-accuracy requirement. The test shall be repeated five times, and th average length of time recorded

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for all measurements shall be considered as the response time of each instrument. (See 3.9).

4.5.6 Power loss. - Power loss shall be measured at end-scale deflection and recorded in terms of current, voltage, voltamperes, or ohms, as applicable. (See 3.10).

4.5.7 Temperature influence. - Analyzers shall be tested at temperatures 20°C. above and below the reference temperature. Readings shall be taken at not less than four points on the scale, of each instrument one of which is at or near the maximum scale value, and the temperature influence computed. The temperature influence shall be considered one-half the percentage change in indication cause by the 20°C. departures from the reference temperature (see 3.11).

4.5.8 Frequency influence. - Analyzers shall be calibrated at nominal frequency, then at frequencies 10 percent above and below the nominal frequency. The change in indication of each instrument shall not exceed the values specified. (see 3.18).

4.5.9 Extreme temperatures. - Analyzers shall be subjected to withstand a temperature of ±65C (±149F) for a period of 16 hours, each instrument shall be capable of indicating freely. When instruments are subjected to such a temperature, the change in indication throughout the scale of each instrument shall not exceed five times the allowance for external temperature influence specified on the applicable detailed requirement sheet. Following subjection of instruments to ±65C (±149F) for 16 hours they shall be returned to ±25C (±77F) where they shall be maintained for 8 hours; and then the calibration shall be checked throughout the scale. The change in indication, due to the effect of heat at ±65C (±149F), shall not exceed an amount equivalent to the rated accuracy.

4.5.10 Overload. -

4.5.10.1 Momentary. - Current circuits shall be subjected to 10 applications of an overload current equal to 10 times end-scale value for a duration of one-half second, each with a one minute interval between successive applications. For instruments provided with a voltage circuit, normal voltage shall be applied to the instruments for the entire period. One hour after the test specified above the pointer shall be reset to zero and the calibration checked over the length of the scale. The change in indication shall not exceed an amount equal to the rated accuracy at any point on the scale.

4.5.10.2 Sustained. - All circuits of instruments shall be capable of withstanding an applied load of 20 percent greater than rated capacity for a period of eight hours. Where a maximum rating is given, this value shall be applied rather than 20 percent greater than rated capacity. Immediately upon disconnecting the instrument at the conclusion of the eight-hour period, the temporary zero shift shall not exceed an amount equal to 1 percent of scale length. Sixteen hours after disconnecting the instrument, and after resetting zero, the permanent error in indication shall not exceed an amount equal to one-half the rated accuracy. Except for the above permissible errors, no permanent damage shall occur as a result of this test.

4.5.11 Humidity resistance. - The analyzer shall be capable of withstanding the humidity test given below:

(a) Two instruments with covers in place and two with covers removed shall be mounted in their normal operating position in a humidity chamber. The chamber shall be maintained for six hours at a temperature of approximately +65 (+149F) and a relative humidity of approximately 95 percent with the instruments energized at two-thirds full-scale deflection. At the end of this time the heating means shall be shut off, the instruments de-energized and allowed to cool for 18 hours in this chamber and atmosphere. During this time the humidity rises as the temperature decreases. Precautions shall be taken to prevent dripping on the instruments. At the end of this time the instruments shall be removed from the chamber and maintained approximately +25C (+77F) for 24 hours. At the completion of this period the instrument shall indicate freely, and any permanent change in indication due to the effect of this test shall not exceed 3 percent of scale length.

4.5.12 Dielectric stress. - The analyzer shall be subjected to a dielectric test voltage of the applicable rms values specified in 3.18. Alternating voltage of 60 cycle frequency and approximating as closely as possible a true sine wave shall be applied continuously for a period of 60 seconds. The measurement of the test voltage shall be made by the voltmeter method, whereby the instrument derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer or through an auxiliary ratio transformer. The test shall be made using high-potential test prods having not less than one-eight-inch radius tips. Any damage or flashover shall be noted.

4.5.13 Insulation resistance. - Insulation-resistance tests shall be made using a potential of 500 volts, direct current; these tests shall precede the dielectric strength tests. The insulation resistance shall be as specified in 3.17.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. -

5.1.1 The equipment shall be preserved and packaged by level A or C; packed by level A, B or C and marked, as specified. (see 6.2); in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended use. - The analyzer is intended to be used in the analysis of electric circuit characteristics at various points on the circuit including those points when motor loads are connected.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preservation and packaging requirement (5.1)
(c) That when a manufacturer can present proof of a previous approval of identical analyzer, preproduction inspection may be waived for the next 1000 analyzers or a period of 5 years, whichever occurs first. After the period, the manufacturer will be required to submit a preproduction sample for test.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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